

Tong QFT — Problem Sheet 3

D. Tong, Quantum Field Theory, Problem Sheet 3 (Dirac fields).

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Quantum Field Theory

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1. Weyl and Dirac Representations of the Clifford Algebra, Q1

The Weyl representation of the Clifford algebra is given by

$$\gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (1.1)$$

Show that these indeed satisfy $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, where $\mathbf{1}$ comes with an implicit 4×4 unit matrix. Find a unitary matrix U such that $(\gamma')^\mu = U\gamma^\mu U^\dagger$, where $(\gamma')^\mu$ form the Dirac representation of the Clifford algebra

$$(\gamma')^0 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad (\gamma')^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}. \quad (1.2)$$

Solution. Write $\sigma^\mu = (\mathbf{1}, \sigma)$, and $\bar{\sigma}^\mu = (\mathbf{1}, -\sigma)$. The anticommutator is given by

$$\begin{aligned} \{\gamma^\mu, \gamma^\nu\} &= \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} + \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \\ &= \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu & 0 \\ 0 & \bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu \end{pmatrix} \\ &= 2\eta^{\mu\nu} \mathbf{1}. \end{aligned} \quad (1.3)$$

U diagonalise γ^0 , and left the rest unchanged. Therefore, we find

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & -\mathbf{1} \end{pmatrix}. \quad (1.4)$$

2. The Lorentz Algebra from Gamma Matrices, Q2

Show that if $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, then

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda. \quad (2.1)$$

Show further that $S^{\mu\nu} \equiv \frac{1}{4}[\gamma^\mu, \gamma^\nu] = \frac{1}{2}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu)$. Use this to confirm that the matrices $S^{\mu\nu}$ form a representation of the Lie algebra of the Lorentz group.

Solution. Note that

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = \gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu - \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda. \quad (2.2)$$

Then, Hence, we find

$$\begin{aligned}\gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - \gamma^\kappa \gamma^\mu \gamma^\lambda \gamma^\nu \\ &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\lambda\nu} \gamma^\kappa \gamma^\mu + \gamma^\kappa \gamma^\mu \gamma^\nu \gamma^\lambda \\ &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\lambda\nu} \gamma^\kappa \gamma^\mu + 2\eta^{\kappa\mu} \gamma^\nu \gamma^\lambda - 2\eta^{\lambda\nu} \gamma^\mu \gamma^\lambda + \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda.\end{aligned}\tag{2.3}$$

Hence, we find

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda.\tag{2.4}$$

Similarly,

$$\begin{aligned}S^{\mu\nu} &= \frac{1}{4}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \\ &= \frac{1}{4}(2\gamma^\mu \gamma^\nu - 2\eta^{\mu\nu}) \\ &= \frac{1}{2}(\gamma^\mu \gamma^\nu - \eta^{\mu\nu}).\end{aligned}\tag{2.5}$$

The Lie algebra is given by

$$\begin{aligned}[S^{\kappa\lambda}, S^{\mu\nu}] &= \frac{1}{4}(\gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu - \eta^{\kappa\lambda} \gamma^\mu \gamma^\nu - \eta^{\mu\nu} \gamma^\kappa \gamma^\lambda - \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda + \eta^{\kappa\lambda} \gamma^\mu \gamma^\nu + \eta^{\mu\nu} \gamma^\kappa \gamma^\lambda) \\ &= \frac{1}{4}[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] \\ &= \frac{1}{4}(2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda) \\ &= \frac{1}{2}(\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - \eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + \eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - \eta^{\kappa\nu} \gamma^\mu \gamma^\lambda) \\ &= \eta^{\lambda\mu} (S^{\kappa\nu} + \frac{1}{2}\eta^{\kappa\nu}) - \eta^{\kappa\mu} (S^{\lambda\nu} + \frac{1}{2}\eta^{\lambda\nu}) + \eta^{\lambda\nu} (S^{\mu\kappa} + \frac{1}{2}\eta^{\mu\kappa}) - \eta^{\kappa\nu} (S^{\mu\lambda} + \frac{1}{2}\eta^{\mu\lambda}) \\ &= \eta^{\lambda\mu} S^{\kappa\nu} - \eta^{\kappa\mu} S^{\lambda\nu} + \eta^{\lambda\nu} S^{\mu\kappa} - \eta^{\kappa\nu} S^{\mu\lambda}.\end{aligned}\tag{2.6}$$

3. Trace Technology and Gamma Matrix Identities, Q3

Using just the algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ (i.e. without resorting to a particular representation), and defining $\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3$, $\not{p} = p_\mu \gamma^\mu$ and $S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$, prove the following results: (Some useful tricks include the cyclicity of the trace, and inserting $(\gamma^5)^2 = 1$ into a trace).

- i. $\text{Tr } \gamma^\mu = 0$
- ii. $\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu}$
- iii. $\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho) = 0$
- iv. $(\gamma^5)^2 = 1$
- v. $\text{Tr } \gamma^5 = 0$
- vi. $\not{p}\not{q} = 2p \cdot q - \not{q}\not{p} = p \cdot q + 2S^{\mu\nu} p_\mu q_\nu$
- vii. $\text{Tr}(\not{p}\not{q}) = 4p \cdot q$
- viii. $\text{Tr}(\not{p}_1 \cdots \not{p}_n) = 0$ if n is odd
- ix. $\text{Tr}(\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = 4[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(p_2 \cdot p_4)]$
- x. $\text{Tr}(\gamma^5 \not{p}_1 \not{p}_2) = 0$
- xi. $\gamma_\mu \not{p} \gamma^\mu = -2\not{p}$

xii. $\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu = 4p_1 \cdot p_2$

xiii. $\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu = -2\not{p}_3 \not{p}_2 \not{p}_1$

xiv. $\text{Tr}(\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = 4i \epsilon_{\mu\nu\rho\sigma} p_1^\mu p_2^\nu p_3^\rho p_4^\sigma$

Solution. Let us consider spinors in d -dimensions.

i. Trivial.

ii.

$$\text{Tr}\{\gamma^\mu \gamma^\nu\} = 2\eta^{\mu\nu} \text{Tr}\{\mathbf{1}\} - \text{Tr}\{\gamma^\nu \gamma^\mu\} \implies \text{Tr}\{\gamma^\mu \gamma^\nu\} = d\eta^{\mu\nu}. \quad (3.1)$$

iii.

$$\text{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho\} = -\text{Tr}\{\gamma^\nu \gamma^\mu \gamma^\rho\} = \text{Tr}\{\gamma^\nu \gamma^\rho \gamma^\mu\} = -\text{Tr}\{\gamma^\rho \gamma^\nu \gamma^\mu\} = 0. \quad (3.2)$$

iv.

$$(\gamma^5)^2 = -\gamma^0 \gamma^1 \gamma^2 \gamma^3 \gamma^0 \gamma^1 \gamma^2 \gamma^3 = -\gamma^1 \gamma^2 \gamma^3 \gamma^1 \gamma^2 \gamma^3 = -\gamma^2 \gamma^3 \gamma^2 \gamma^3 = \mathbf{1}. \quad (3.3)$$

v. Will turn out to be trivial using the general trace of for γ -matrices.

vi.

$$\not{p} \not{q} = p_\mu q_\nu (2\eta^{\mu\nu} - \gamma^\nu \gamma^\mu) = 2p \cdot q - \not{q} \not{p} = p_\mu q_\nu (\eta^{\mu\nu} + 2S^{\mu\nu}) = p \cdot q + 2S^{\mu\nu} p_\mu q_\nu. \quad (3.4)$$

vii.

$$\text{Tr}\{\not{p} \not{q}\} = d p \cdot q. \quad (3.5)$$

viii. Follows from iii by induction.

ix.

$$\begin{aligned} \text{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\lambda\} &= 2(\eta^{\mu\nu} \text{Tr}\{\gamma^\rho \gamma^\lambda\} - \eta^{\mu\rho} \text{Tr}\{\gamma^\nu \gamma^\lambda\} + \eta^{\mu\lambda} \text{Tr}\{\gamma^\nu \gamma^\rho\}) - \text{Tr}\{\gamma^\nu \gamma^\rho \gamma^\lambda \gamma^\mu\} \\ \implies \text{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\lambda\} &= d(\eta^{\mu\nu} \eta^{\rho\lambda} - \eta^{\mu\rho} \eta^{\nu\lambda} + \eta^{\mu\lambda} \eta^{\nu\rho}) \\ \implies \text{Tr}\{\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\} &= d[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(p_2 \cdot p_4)]. \end{aligned} \quad (3.6)$$

x. A cleaner argument: by anticommutation of γ^5 with all γ^μ , the trace is totally antisymmetric in (μ, ν) . The only available rank-2 tensor in 4d is $\eta^{\mu\nu}$, which is symmetric. Hence the trace must vanish:

$$\text{Tr}\{\gamma^5 \not{p}_1 \not{p}_2\} = 0. \quad (3.7)$$

xi. Insert $\eta_{\mu\nu} \gamma^\mu \gamma^\nu = d \mathbf{1}$ and use $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$:

$$\gamma_\mu \not{p} \gamma^\mu = p_\nu \gamma_\mu \gamma^\nu \gamma^\mu = p_\nu (2\gamma_\mu^\nu - \gamma^\nu \gamma_\mu) \gamma^\mu = 2\not{p} - d\not{p} = (2-d)\not{p} = -2\not{p}. \quad (3.8)$$

xii. Apply the previous result twice. Using $\gamma_\mu \gamma^\nu \gamma^\rho \gamma^\mu = 4\eta^{\nu\rho}$ (proved similarly):

$$\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu = p_{1\nu} p_{2\rho} \gamma_\mu \gamma^\nu \gamma^\rho \gamma^\mu = 4p_{1\nu} p_{2\rho} \eta^{\nu\rho} = 4p_1 \cdot p_2. \quad (3.9)$$

xiii. Similarly $\gamma_\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu$, hence

$$\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu = -2\not{p}_3 \not{p}_2 \not{p}_1. \quad (3.10)$$

xiv. By the same antisymmetry argument as in (x), the trace $\text{Tr}\{\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\}$ must be totally antisymmetric in the four momentum indices. The only such rank-4 tensor in 4d is $\epsilon^{\mu\nu\rho\sigma}$, so

$$\text{Tr}\{\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\} = c \epsilon^{\mu\nu\rho\sigma}. \quad (3.11)$$

The constant c is fixed by setting $(\mu, \nu, \rho, \sigma) = (0, 1, 2, 3)$:

$$\text{Tr}\{\gamma^5 \gamma^0 \gamma^1 \gamma^2 \gamma^3\} = i \text{Tr}\{\gamma^5 \gamma^5\} = 4i, \quad (3.12)$$

using $\gamma^5 = -i\gamma^0 \gamma^1 \gamma^2 \gamma^3$ and $(\gamma^5)^2 = \mathbf{1}$. With $\epsilon^{0123} = -\epsilon_{0123} = -1$, we find $c = -4i$, giving

$$\text{Tr}\{\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\} = -4i \epsilon_{\mu\nu\rho\sigma} p_1^\mu p_2^\nu p_3^\rho p_4^\sigma. \quad (3.13)$$

4. Plane-Wave Spinor Solutions and Orthogonality, Q4

The plane-wave solutions to the Dirac equation are

$$u^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \quad \text{and} \quad v^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ -\sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \quad (4.1)$$

where $\sigma^\mu = (1, \sigma^i)$ and $\bar{\sigma}^\mu = (1, -\sigma^i)$ and ξ^s , with $s = 1, 2$, is a basis of orthonormal two-component spinors, satisfying $(\xi^r)^\dagger \cdot \xi^s = \delta^{rs}$. Show that

$$u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) = 2p_0 \delta^{rs} \quad (4.2)$$

$$\bar{u}^r(\mathbf{p}) \cdot u^s(\mathbf{p}) = 2m \delta^{rs} \quad (4.3)$$

and similarly,

$$v^r(\mathbf{p})^\dagger \cdot v^s(\mathbf{p}) = 2p_0 \delta^{rs} \quad (4.4)$$

$$\bar{v}^r(\mathbf{p}) \cdot v^s(\mathbf{p}) = -2m \delta^{rs} \quad (4.5)$$

Show also that the orthonality condition between u and v is

$$\bar{u}^s(\mathbf{p}) \cdot v^r(\mathbf{p}) = 0 \quad (4.6)$$

while taking the inner product using $\dagger$ requires an extra minus sign

$$u^r(\mathbf{p})^\dagger \cdot v^s(-\mathbf{p}) = 0. \quad (4.7)$$

Solution.

$$\begin{aligned} u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) &= \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^{\dagger r} & \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \\ &= \delta^{rs} \left[\sqrt{(p \cdot \sigma)(p \cdot \sigma^\dagger)} + \sqrt{(p \cdot \bar{\sigma})(p \cdot \bar{\sigma}^\dagger)} \right] \\ &= 2\delta^{rs} \sqrt{(p \cdot \sigma)(p \cdot \sigma^\dagger)}, \end{aligned} \quad (4.8)$$

Since the Pauli matrices are Hermitian, we find

$$u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) = 2\delta^{rs} (p \cdot \sigma + p \cdot \bar{\sigma}) = 2p_0 \delta^{rs}. \quad (4.9)$$

Similarly, we find

$$\begin{aligned} \bar{u}^r(\mathbf{p}) \cdot u^s(\mathbf{p}) &= \begin{pmatrix} \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} & \sqrt{p \cdot \sigma} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \\ &= \delta^{rs} \left[\sqrt{(p \cdot \sigma)(p \cdot \bar{\sigma}^\dagger)} + \sqrt{(p \cdot \bar{\sigma})(p \cdot \sigma^\dagger)} \right] \\ &= 2\delta^{rs} \sqrt{(p \cdot \sigma)(p \cdot \bar{\sigma})} \\ &= 2m \delta^{rs}, \end{aligned} \quad (4.10)$$

where we have expanded $p \cdot \sigma$ and $p \cdot \bar{\sigma}$ and evaluated them explicitly.

Similarly, one can easily obtain

$$v^r(\mathbf{p})^\dagger \cdot v^s(\mathbf{p}) = 2p_0 \delta^{rs} \quad (4.11)$$

$$\bar{v}^r(\mathbf{p}) \cdot v^s(\mathbf{p}) = -2m \delta^{rs}. \quad (4.12)$$

Now, we demonstrate the orthogonality condition,

$$\bar{u}^s(\mathbf{p}) \cdot v^r(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger s} & \sqrt{p \cdot \sigma} \xi^{\dagger s} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^r \\ -\sqrt{p \cdot \bar{\sigma}} \xi^r \end{pmatrix} = 0, \quad (4.13)$$

by Hermiticity of Pauli matrices. Similarly,

$$u^r(\mathbf{p})^\dagger \cdot v^s(-\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^{\dagger r} & \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{\tilde{p} \cdot \sigma} \xi^s \\ -\sqrt{\tilde{p} \cdot \bar{\sigma}} \xi^s \end{pmatrix}, \quad (4.14)$$

where $\tilde{p}^\mu = (p_0, -\mathbf{p})$. Now, note that $\tilde{p} \cdot \sigma = p \cdot \bar{\sigma}$, we see that the above inner product vanishes.

5. Spin Sums and Completeness Relation, Q5

Using the same notation as Question 4 show that

$$\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) = \not{p} + m \quad (5.1)$$

$$\sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) = \not{p} - m \quad (5.2)$$

where, rather than being contracted, the two spinors on the left-hand side are placed back to back to form a 4×4 matrix.

Solution.

$$\begin{aligned} \sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) &= \sum_{s=1}^2 \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger s} & \sqrt{p \cdot \sigma} \xi^{\dagger s} \end{pmatrix} \\ &= \begin{pmatrix} m & p \cdot \sigma \\ p \cdot \bar{\sigma} & m \end{pmatrix} \sum_{s=1}^2 \xi^s \xi^{\dagger s}. \end{aligned} \quad (5.3)$$

The latter sum can be obtained by considering

$$\sum_{s=1}^2 \delta^{sa} \delta^{sb} = \delta^{ab} \implies \sum_{s=1}^2 (\xi^{\dagger s} \cdot \xi^a) (\xi^{\dagger s} \cdot \xi^b) = \xi^{\dagger b} \cdot \xi^a \implies \sum_{s=1}^2 \xi_\alpha^s \xi_\beta^{\dagger s} = \delta_{\alpha\beta}. \quad (5.4)$$

Therefore, we find

$$\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) = \not{p} + m. \quad (5.5)$$

Almost identically, we find

$$\sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) = \not{p} - m. \quad (5.6)$$

6. Canonical Quantization of the Dirac Field, Q6

The Fourier decomposition of the Dirac operator $\psi(\mathbf{x})$ and the conjugate field $\psi^\dagger(\mathbf{x})$ is given by

$$\psi(\mathbf{x}) = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} [b_{\mathbf{p}}^s u^s(\mathbf{p}) e^{+i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^{s\dagger} v^s(\mathbf{p}) e^{-i\mathbf{p} \cdot \mathbf{x}}] \quad (6.1)$$

$$\psi^\dagger(\mathbf{x}) = \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} [b_{\mathbf{p}}^{s\dagger} u^s(\mathbf{p})^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^s v^s(\mathbf{p})^\dagger e^{+i\mathbf{p} \cdot \mathbf{x}}] \quad (6.2)$$

The creation and annihilation operators are taken to satisfy

$$\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} = (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \quad (6.3)$$

$$\{c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \quad (6.4)$$

with all other anti-commutators vanishing,

$$\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^s\} = \{c_{\mathbf{p}}^r, c_{\mathbf{q}}^s\} = \{b_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = \{b_{\mathbf{p}}^r, c_{\mathbf{q}}^s\} = \dots = 0. \quad (6.5)$$

Show that these imply that the field and its conjugate momenta satisfy the anti-commutation relations,

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta(\mathbf{y})\} = \{\psi_\alpha^\dagger(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = 0 \quad (6.6)$$

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (6.7)$$

(Note: The calculation is very similar to that for the bosonic field, but at some point you will need to make use of the identities (5.1) and (5.2)).

Solution. We compute $\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\}$ directly. Substituting the Fourier decompositions,

$$\begin{aligned} \{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} &= \sum_{r,s} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{q}}}} \\ &\times \left[\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} u_\alpha^r(\mathbf{p}) u_\beta^{s\dagger}(\mathbf{q}) e^{i\mathbf{p}\cdot\mathbf{x} - i\mathbf{q}\cdot\mathbf{y}} \right. \\ &\quad \left. + \{c_{\mathbf{p}}^{r\dagger}, c_{\mathbf{q}}^s\} v_\alpha^r(\mathbf{p}) v_\beta^{s\dagger}(\mathbf{q}) e^{-i\mathbf{p}\cdot\mathbf{x} + i\mathbf{q}\cdot\mathbf{y}} \right] \end{aligned} \quad (6.8)$$

where all cross terms $\{b, c\}$ and $\{b, c^\dagger\}$ vanish. Then,

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \sum_s \left[u_\alpha^s(\mathbf{p}) u_\beta^{s\dagger}(\mathbf{p}) e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + v_\alpha^s(\mathbf{p}) v_\beta^{s\dagger}(\mathbf{p}) e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]. \quad (6.9)$$

Now we use the spin sum identities. Note that $\bar{u} = u^\dagger \gamma^0$ so $u^\dagger = \bar{u} \gamma^0$, giving

$$\sum_s u^s u^{s\dagger} = \sum_s u^s \bar{u}^s \gamma^0 = (\not{p} + m) \gamma^0, \quad \sum_s v^s v^{s\dagger} = (\not{p} - m) \gamma^0. \quad (6.10)$$

Substituting back,

$$\{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left[(\not{p} + m)_{\alpha\dot{\beta}} (\gamma^0)_{\dot{\beta}\beta} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + (\not{p} - m)_{\alpha\dot{\beta}} (\gamma^0)_{\dot{\beta}\beta} e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]. \quad (6.11)$$

Expanding $\not{p} = p_0 \gamma^0 - p_i \gamma^i$ and using $(\gamma^0)^2 = \mathbf{1}$, the m terms cancel between the two exponentials after relabelling $\mathbf{p} \rightarrow -\mathbf{p}$ in the second integral, and the $p_i \gamma^i \gamma^0$ terms also cancel by the same relabelling. What remains is

$$\begin{aligned} \{\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})\} &= \int \frac{d^3p}{(2\pi)^3} \frac{2p_0}{2E_{\mathbf{p}}} (\gamma^0 \gamma^0)_{\alpha\beta} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \\ &= \delta_{\alpha\beta} \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \end{aligned} \quad (6.12)$$

The vanishing of $\{\psi_\alpha(\mathbf{x}), \psi_\beta(\mathbf{y})\}$ follows similarly, since the cross terms $\{b_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = 0$ and the surviving terms $\{b, b\} = \{c^\dagger, c^\dagger\} = 0$ all vanish.

7. The Dirac Hamiltonian, Q7

Using the results of Question 6, show that the quantum Hamiltonian

$$H = \int d^3x \bar{\psi} (-i\gamma^i \partial_i + m) \psi \quad (7.1)$$

can be written, after normal ordering, as

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (7.2)$$

(Note: Again, the calculation is very similar to that for the bosonic field. This time you will need to make use of the identities derived in Questions 4 and 5).

Solution. Substituting the Fourier decompositions and using $\bar{\psi} = \psi^\dagger \gamma^0$,

$$H = \sum_{r,s} \int d^3x \int \frac{d^3p}{(2\pi)^3} \frac{E_{\mathbf{p}}}{\sqrt{2E_{\mathbf{p}}}} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{q}}}} \times [b_{\mathbf{q}}^{r\dagger} u^r(\mathbf{q})^\dagger e^{-i\mathbf{q}\cdot\mathbf{x}} + c_{\mathbf{q}}^r v^r(\mathbf{q})^\dagger e^{+i\mathbf{q}\cdot\mathbf{x}}] [b_{\mathbf{p}}^s u^s(\mathbf{p}) e^{+i\mathbf{p}\cdot\mathbf{x}} - c_{\mathbf{p}}^{s\dagger} v^s(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}}] \quad (7.3)$$

where we applied $(-i\gamma^i \partial_i + m)u^s(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}} = E_{\mathbf{p}}\gamma^0 u^s e^{i\mathbf{p}\cdot\mathbf{x}}$ and $(-i\gamma^i \partial_i + m)v^s e^{-i\mathbf{p}\cdot\mathbf{x}} = -E_{\mathbf{p}}\gamma^0 v^s e^{-i\mathbf{p}\cdot\mathbf{x}}$ from the Dirac equation, and used $(\gamma^0)^2 = \mathbf{1}$ to cancel the γ^0 from $\bar{\psi} = \psi^\dagger \gamma^0$. Integrating over d^3x gives delta functions, setting $\mathbf{q} = \mathbf{p}$ for diagonal terms and $\mathbf{q} = -\mathbf{p}$ for cross terms:

$$H = \sum_{r,s} \int \frac{d^3p}{(2\pi)^3} \frac{E_{\mathbf{p}}}{2E_{\mathbf{p}}} \left[u^r(\mathbf{p})^\dagger u^s(\mathbf{p}) b_{\mathbf{p}}^{r\dagger} b_{\mathbf{p}}^s - v^r(\mathbf{p})^\dagger v^s(\mathbf{p}) c_{\mathbf{p}}^r c_{\mathbf{p}}^{s\dagger} + u^r(-\mathbf{p})^\dagger (-v^s(\mathbf{p})) b_{-\mathbf{p}}^{r\dagger} c_{\mathbf{p}}^{s\dagger} + v^r(-\mathbf{p})^\dagger u^s(\mathbf{p}) c_{-\mathbf{p}}^r b_{\mathbf{p}}^s \right]. \quad (7.4)$$

The cross terms vanish by the orthogonality relation $u^r(-\mathbf{p})^\dagger v^s(\mathbf{p}) = 0$. Applying the normalisation $u^r(\mathbf{p})^\dagger u^s(\mathbf{p}) = v^r(\mathbf{p})^\dagger v^s(\mathbf{p}) = 2E_{\mathbf{p}}\delta^{rs}$ from (5) and (6),

$$H = \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}] = \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left[b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - (-c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0)) \right]. \quad (7.5)$$

After normal ordering, the vacuum contribution $\delta^{(3)}(0)$ is discarded and the sign of the c term flips, giving

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (7.6)$$

Note the crucial sign difference from the bosonic case: here normal ordering the anticommuting $c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}$ flips the sign to give a positive contribution, whereas for bosons it produces an infinite additive constant.

8. The Spin-Statistics Theorem, Q8

The purpose of this question is to give you a glimpse into the spin-statistics theorem. This theorem roughly says that if you try to quantize a field with the wrong statistics, bad things will happen. Here we'll see what goes wrong if you try to quantize a spin 1/2 field as a boson. We start with the usual decomposition (11). This time we choose bosonic commutation relations for the annihilation and creation operators,

$$[b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}] = (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \quad (8.1)$$

$$[c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}] = -(2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \quad (8.2)$$

with all other commutators vanishing. Note the strange minus sign for the c operators. Repeat the calculation of Question 6 to show that these are equivalent to the commutation relations,

$$[\psi_\alpha(\mathbf{x}), \psi_\beta(\mathbf{y})] = [\psi_\alpha^\dagger(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] = 0 \quad (8.3)$$

$$[\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (8.4)$$

Now repeat the calculation of Question 7, to show that, after normal ordering, the Hamiltonian is given by

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (8.5)$$

This Hamiltonian is not bounded below: you can lower the energy indefinitely by creating more and more c particles. This is the reason a theory of bosonic spin 1/2 particles is sick.

Solution. The calculation proceeds exactly as in Questions 6 and 7. The field commutator gives

$$\begin{aligned}
 [\psi_\alpha(\mathbf{x}), \psi_\beta^\dagger(\mathbf{y})] &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \sum_s \left[u_\alpha^s(\mathbf{p}) u_\beta^{s\dagger}(\mathbf{p}) e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} - v_\alpha^s(\mathbf{p}) v_\beta^{s\dagger}(\mathbf{p}) e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \right] \\
 &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left[(\not{p} + m) \gamma^0 e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} - (\not{p} - m) \gamma^0 e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \right]_{\alpha\beta} \\
 &= \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y})
 \end{aligned} \tag{8.6}$$

where the minus sign from $[c, c^\dagger] = -(2\pi)^3 \delta^{(3)}$ cancels the minus sign in the $v\bar{v}$ spin sum, reproducing the same result as the fermionic case. For the Hamiltonian,

$$\begin{aligned}
 H &= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}] \\
 &= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0)].
 \end{aligned} \tag{8.7}$$

Note that now $c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger} = c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0)$ since these are bosonic operators. After normal ordering,

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s] \tag{8.8}$$

where the minus sign arises from the $[c, c^\dagger] = -(2\pi)^3 \delta^{(3)}$ relation flipping the sign under normal ordering.